

Identifying turning points in early adult life

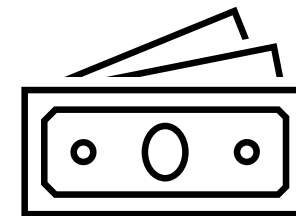
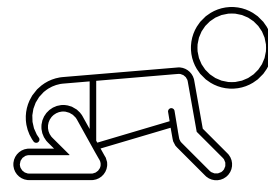
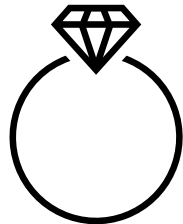
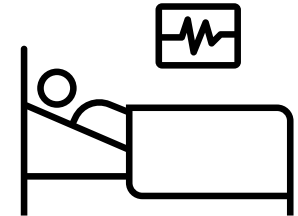
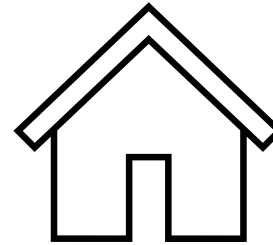
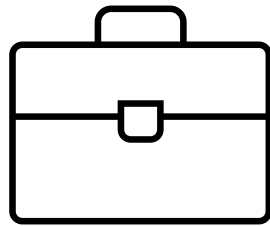
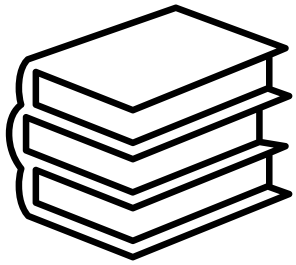
Katie Spoon

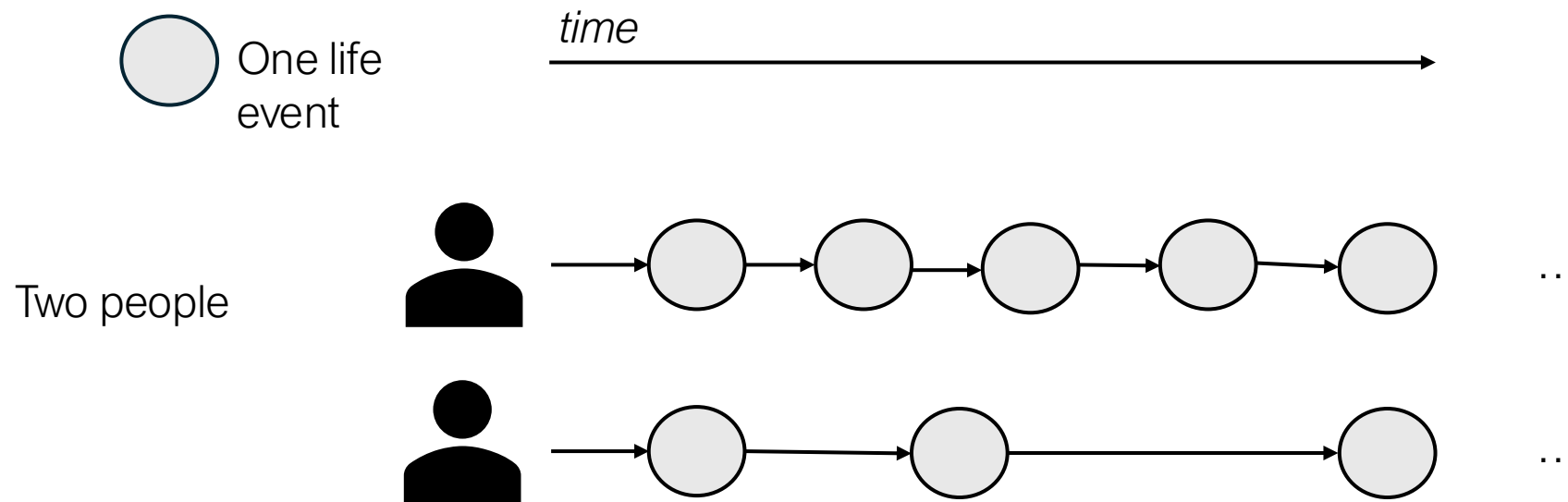
Data Science Fellow and New Map of Life Fellow, Stanford University

July 1, 2026

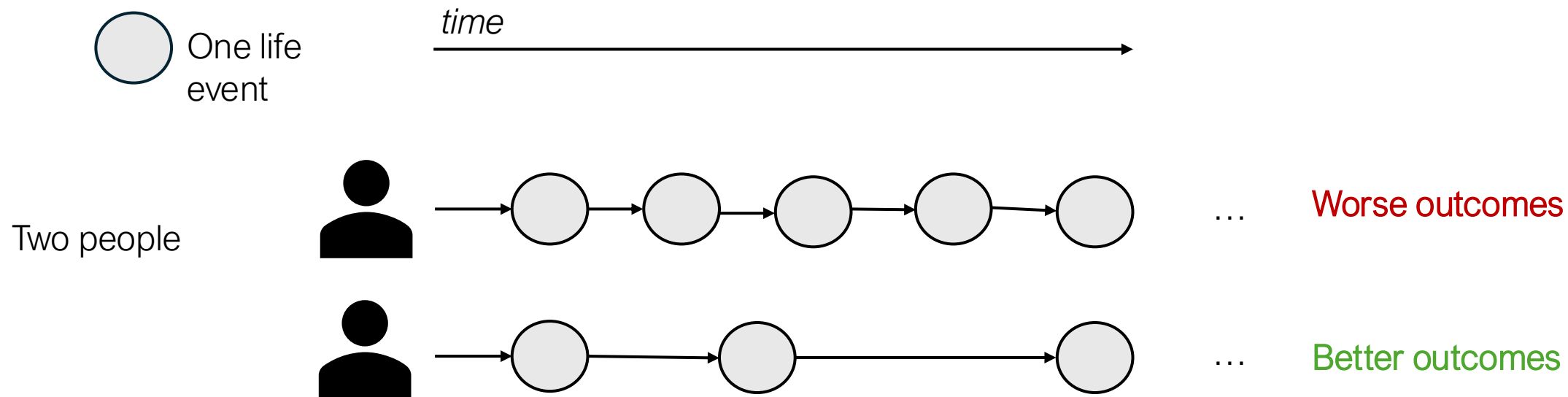
Joint work with Bernardo Mackenna, João Souto Maior, Tyler Matteson, Phil Chodrow, Carolina Chávez-Ruelas, and Carolina Pradier

Early adulthood is a time of change and many consequential life events tend to occur, both positive and negative

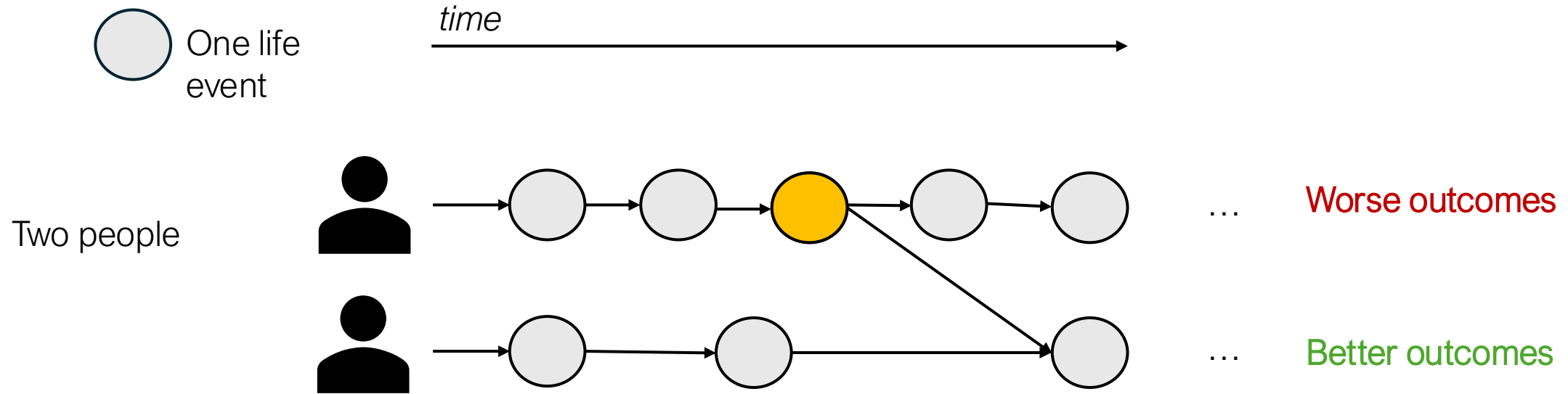




Each person has a life **trajectory**

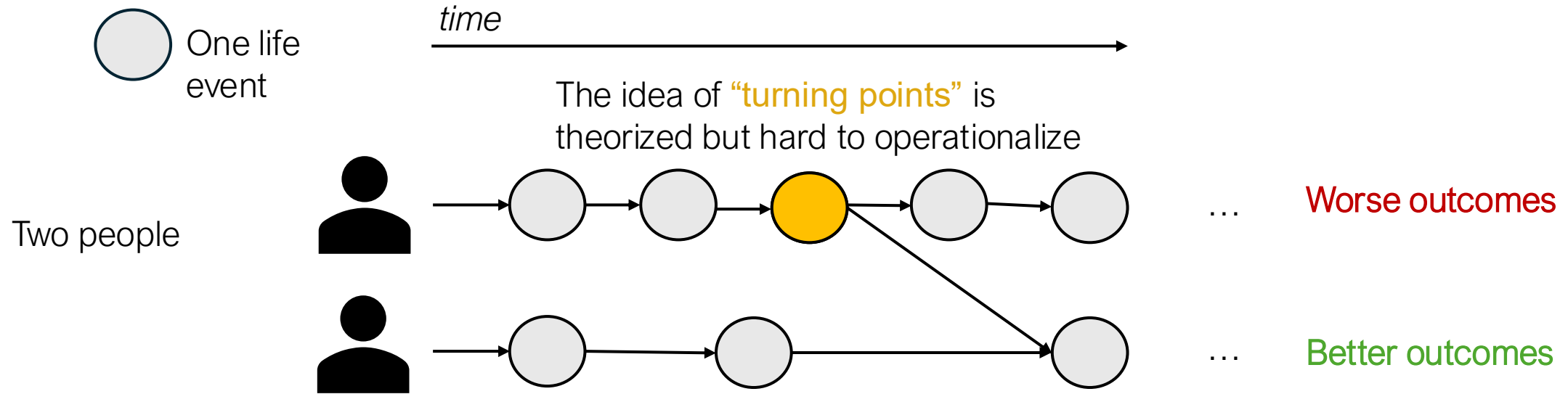


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Are there events that persistently redirect a person's trajectory to one with better outcomes?



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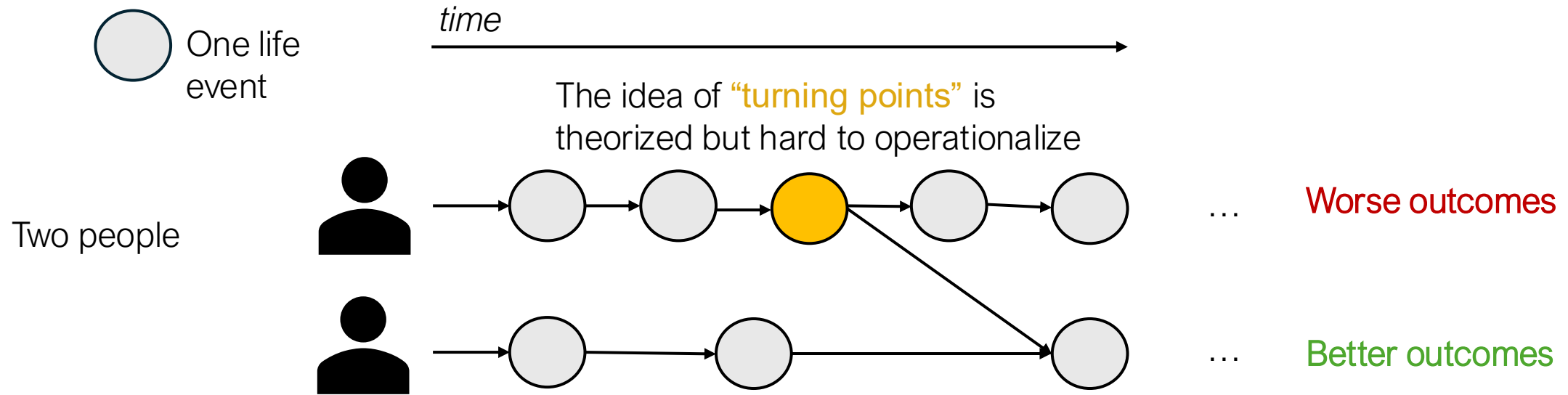
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Elder (1986). Military times and turning points in men's lives. *Developmental Psychology*.

Sampson & Laub (1992). Crime and deviance in the life course. *Annual Review of Sociology*.

Rutter (1996). Transitions and turning points in developmental psychopathology: As applied to the age span between childhood and mid-adulthood. *International Journal of Behavioral Development*.

Gotlib & Wheaton (1997). Stress and adversity over the life course: Trajectories and turning points. *Cambridge University Press*.



Which life events function as turning points in early adult life and *when* do they occur?

National Longitudinal Study of 1972



~20,000 high school seniors



Representative U.S. sample



1972-1986; five follow-ups



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Representative U.S. sample



1972-1986; five follow-ups

5,000 columns!
Very rich data, but that
richness is the
methodological challenge

B. When did you attend this school? (CIRCLE THE FIRST AND LAST MONTHS FOR EACH TIME PERIOD AT THIS SCHOOL. DRAW A LINE BETWEEN THE CIRCLED DATES.)

	JAN	FEB	MAR	APR	MAY	JUNE	JUL	AUG	SEPT	OCT	NOV	DEC
1979	--	--	--	--	--	--	--	--	--	10	11	12
1980	01	02	03	04	05	06	07	08	09	10	11	12
1981	01	02	03	04	05	06	07	08	09	10	11	12
1982	01	02	03	04	05	06	07	08	09	10	11	12
1983	01	02	03	04	05	06	07	08	09	10	11	12
1984	01	02	03	04	05	06	07	08	09	10	11	12
1985	01	02	03	04	05	06	07	08	09	10	11	12
1986	01	02	03	04	05	06	07	--	--	--	--	--

Example

JAN	FEB	MAR	...	NOV	DEC
01	02	03	...	11	12
01	02	03	...	11	12
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Turning points are a central concept in life course theory, but operationalizing them, especially given the richness of the data, requires first choosing how to represent trajectories.

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State-based
representations treat life
as occupying categories or
states over time

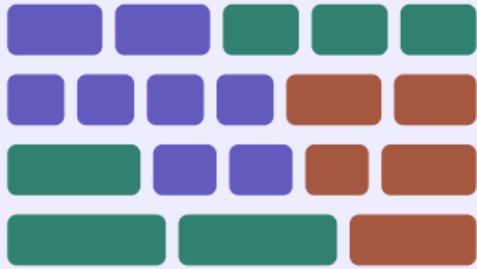
Enrolled
in college

Employed

Enrolled in
grad school

Employed

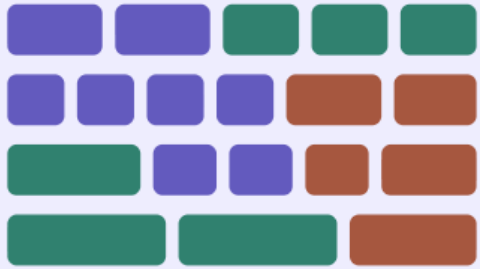
Sequence analysis of states



Traditional approach: very interpretable, but doesn't scale well

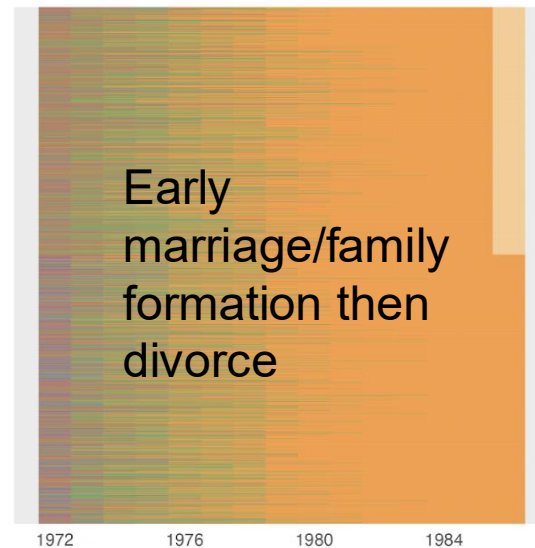
As a result, states must be pre-specified, limiting discovery to what you'd expect to find

Sequence analysis of states

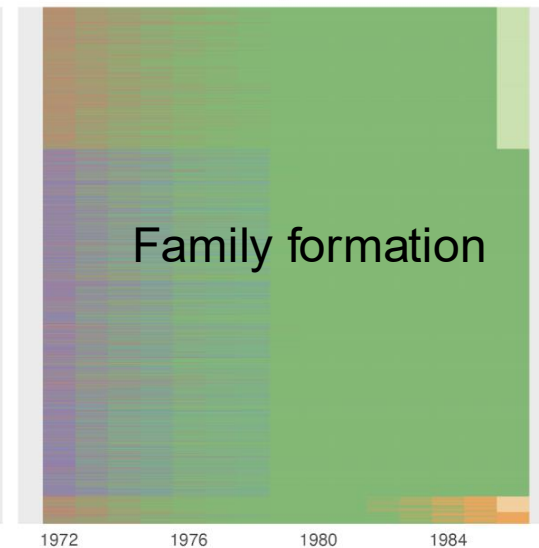


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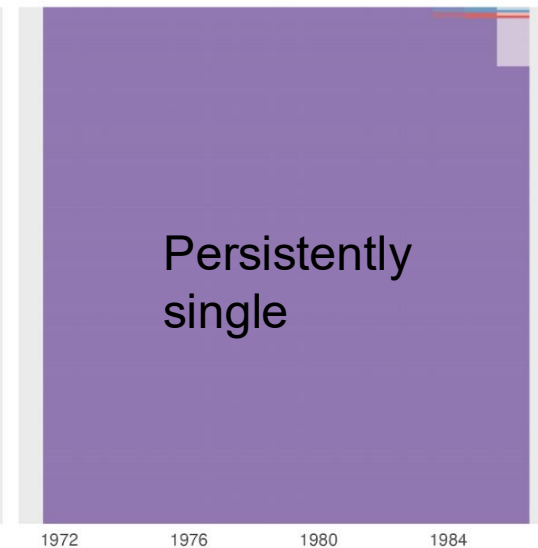
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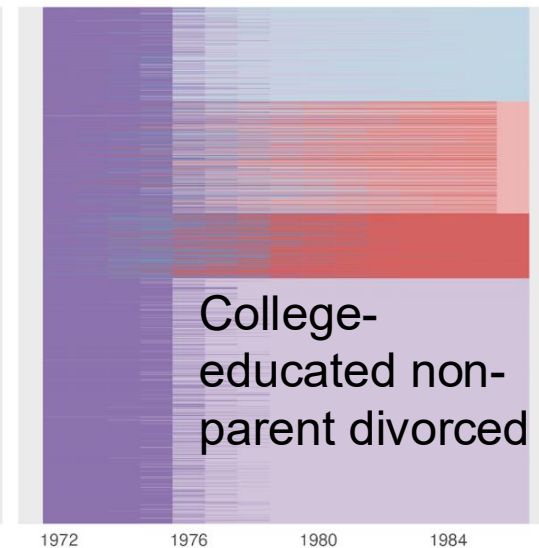
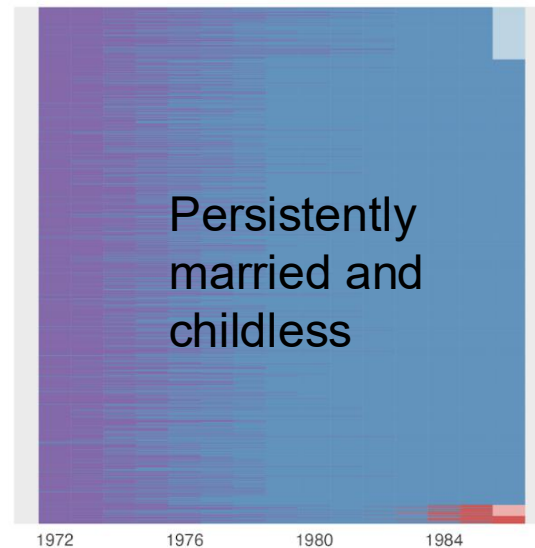
4



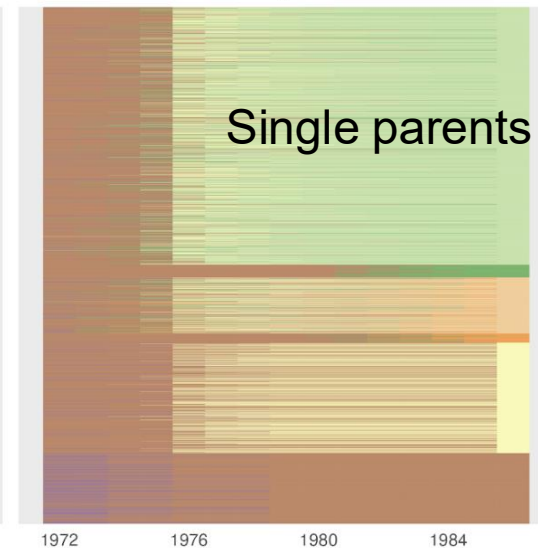
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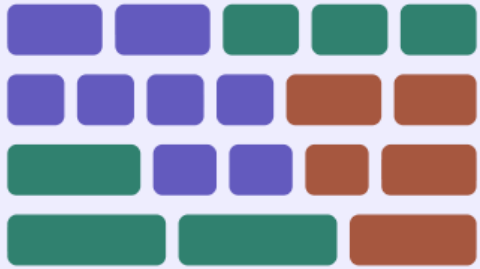
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+

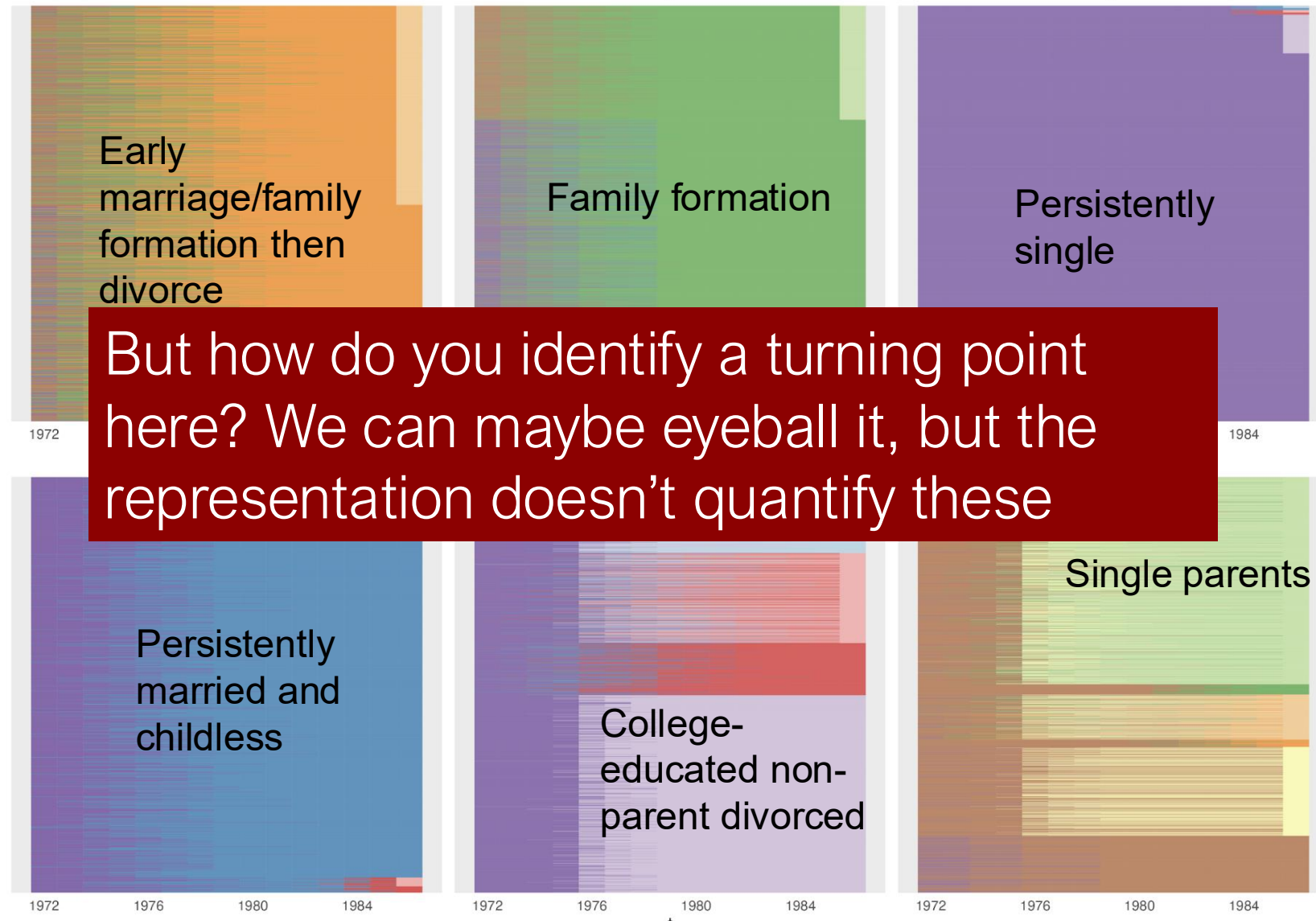


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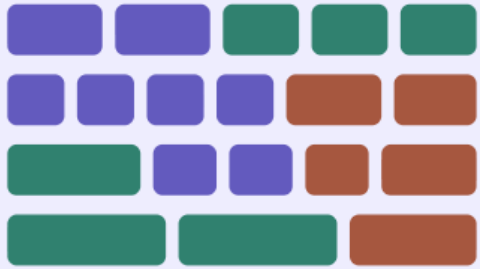


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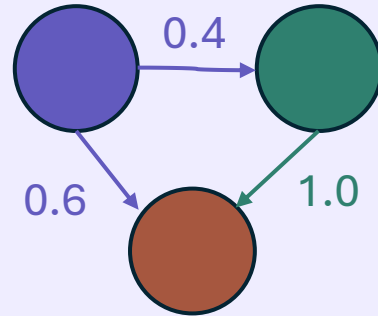
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Markov models of states



Estimates how likely it is to move between **states**; can cluster to find groups of states where trajectories circulate

But has a “memoryless” assumption, so timing information is lost

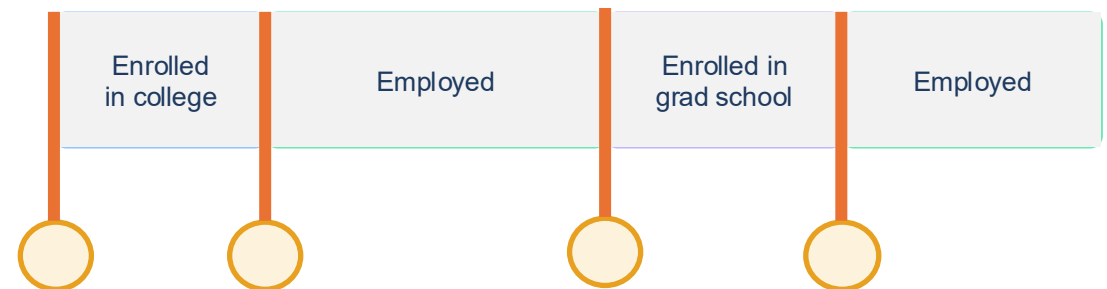
Marriage in 1973 and marriage in 1986 are identical in this model

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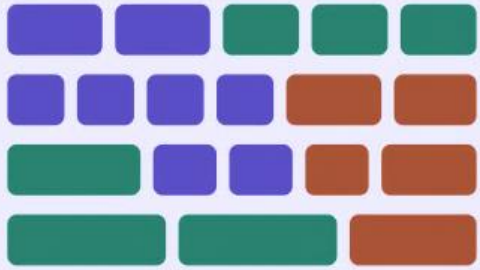
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over time



Transition-based
representations treat life
changes as the unit of
analysis



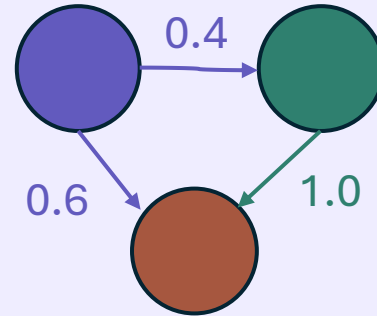
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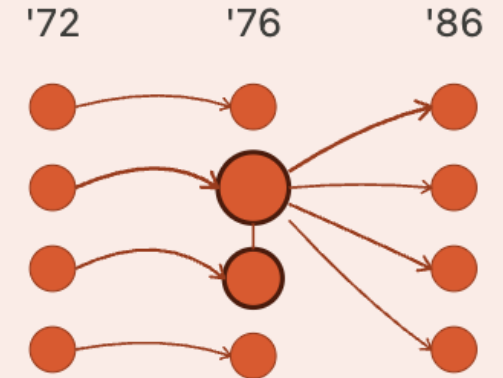


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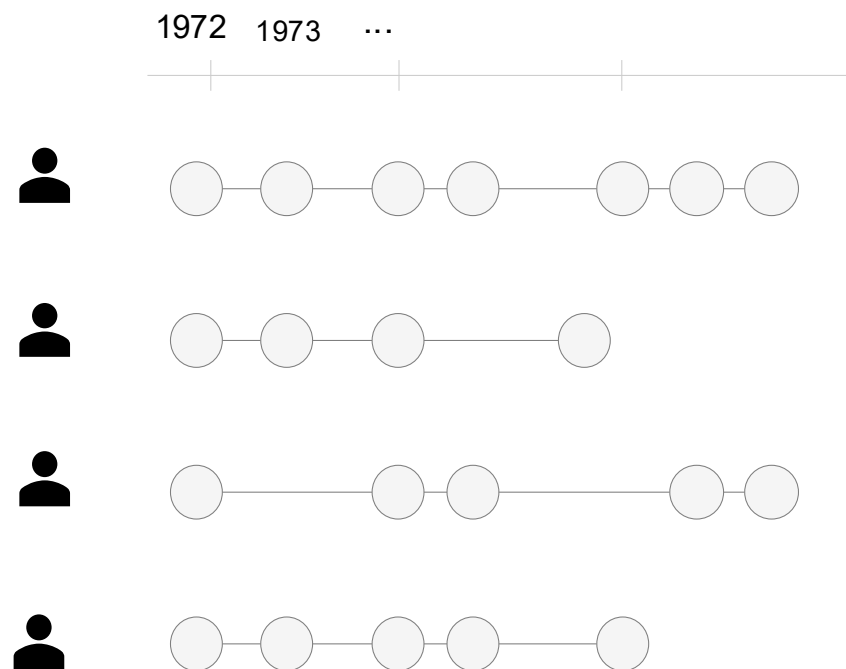
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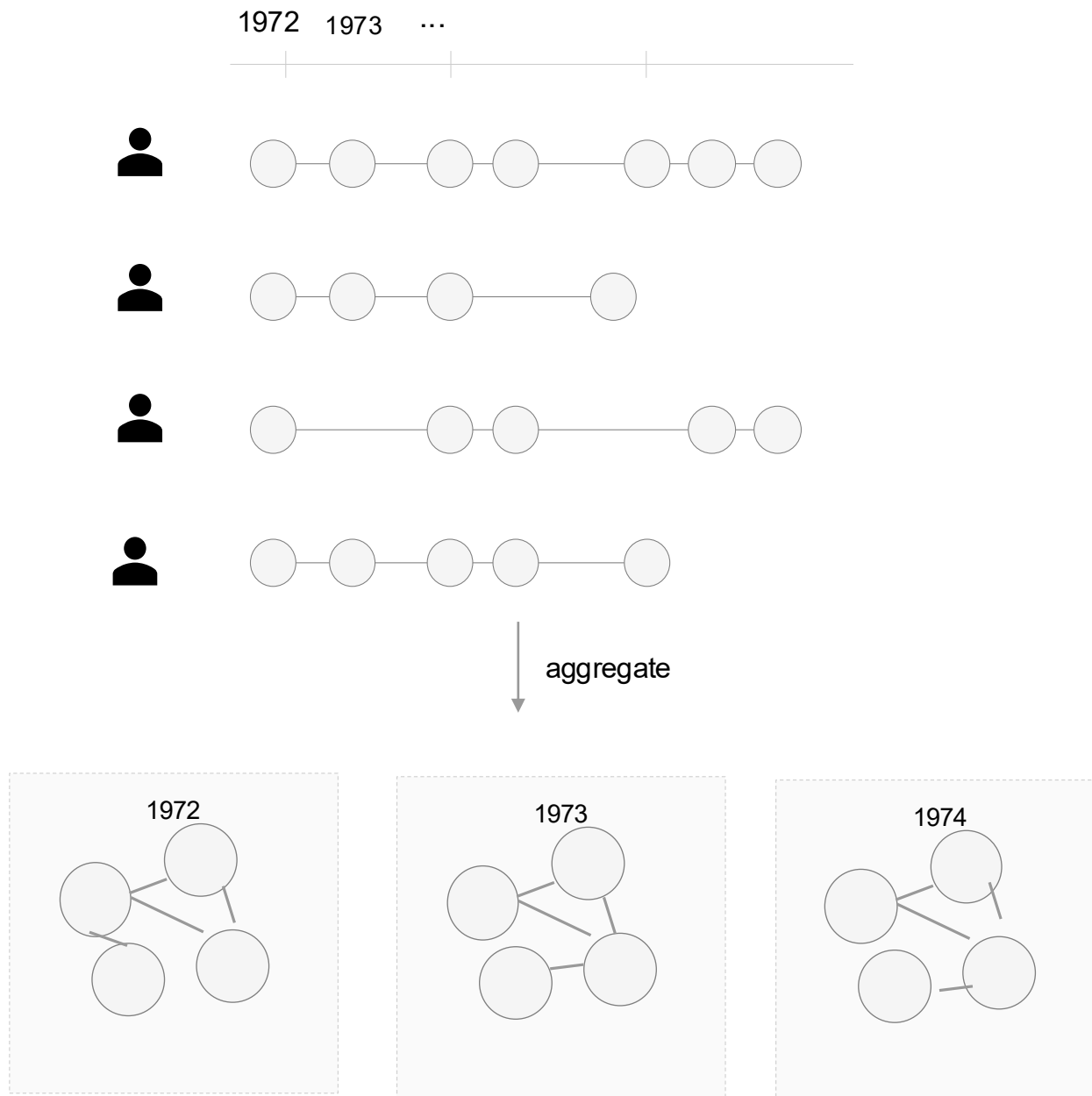
Temporal networks of transitions



Surfaces both **which** events are structurally important and **when** they are most important

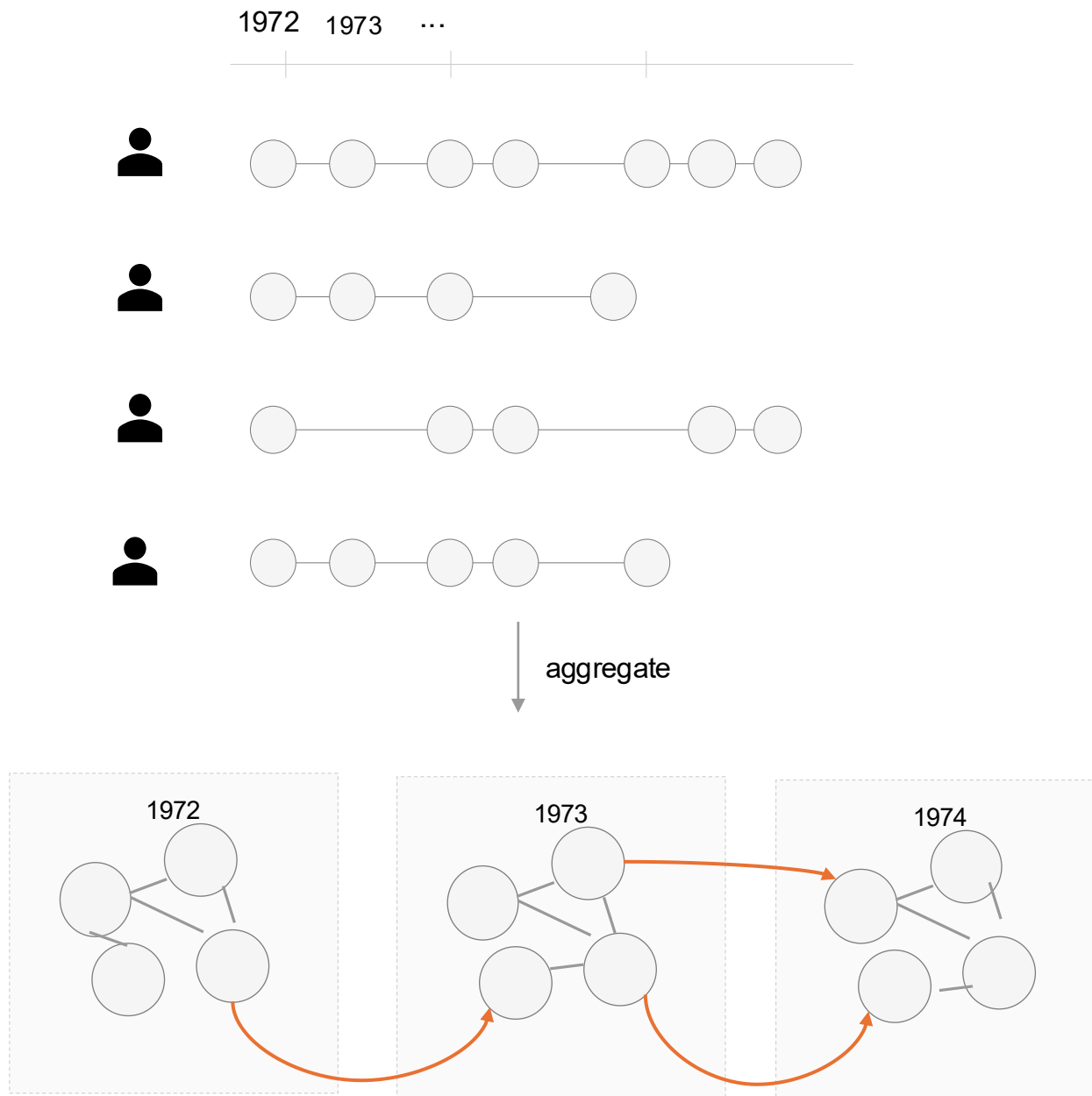


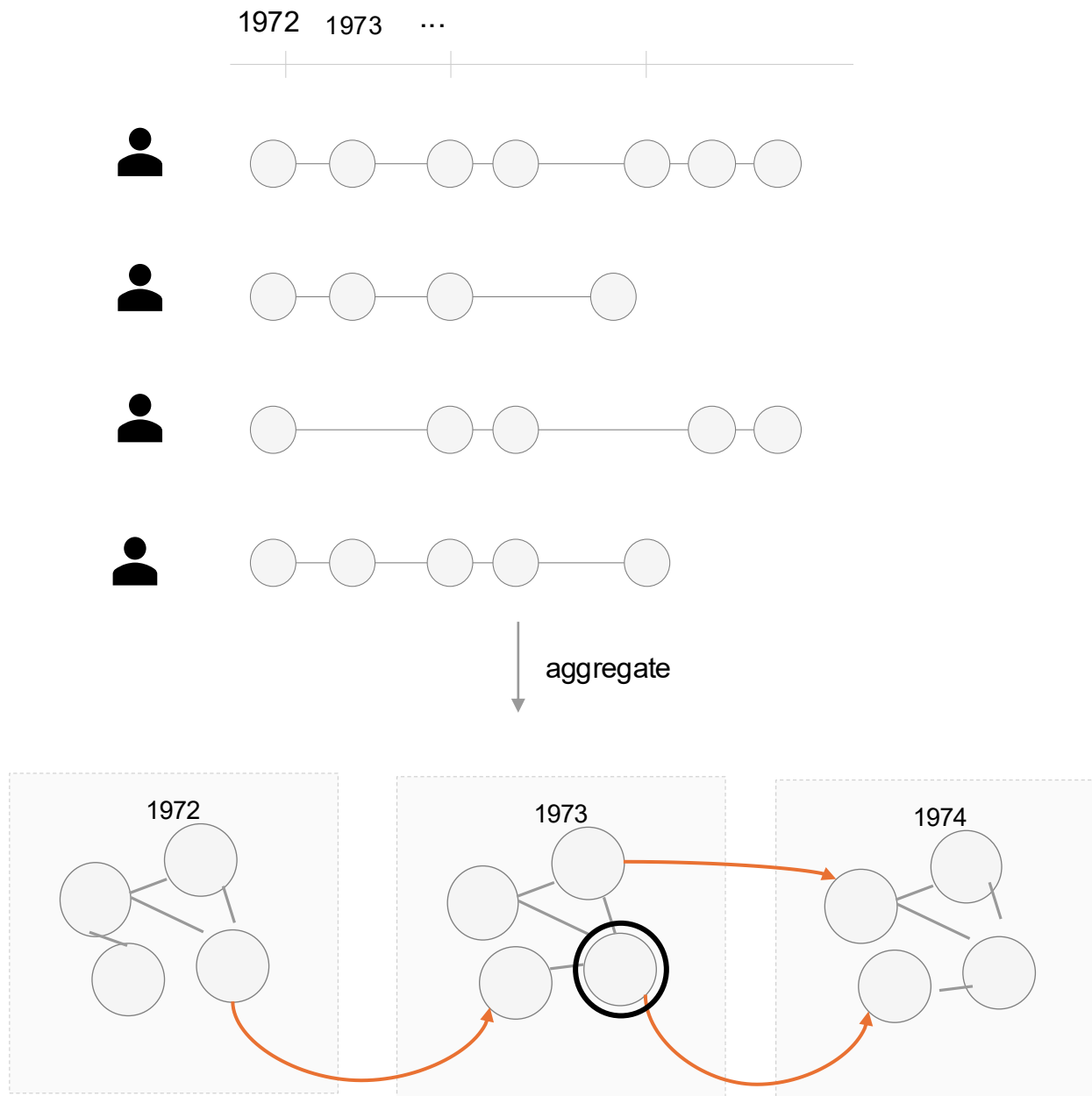
**Each person has a
sequence of life events
over 14 years**



**Each person has a
sequence of life events
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Lines within bins are
undirected co-occurrence.



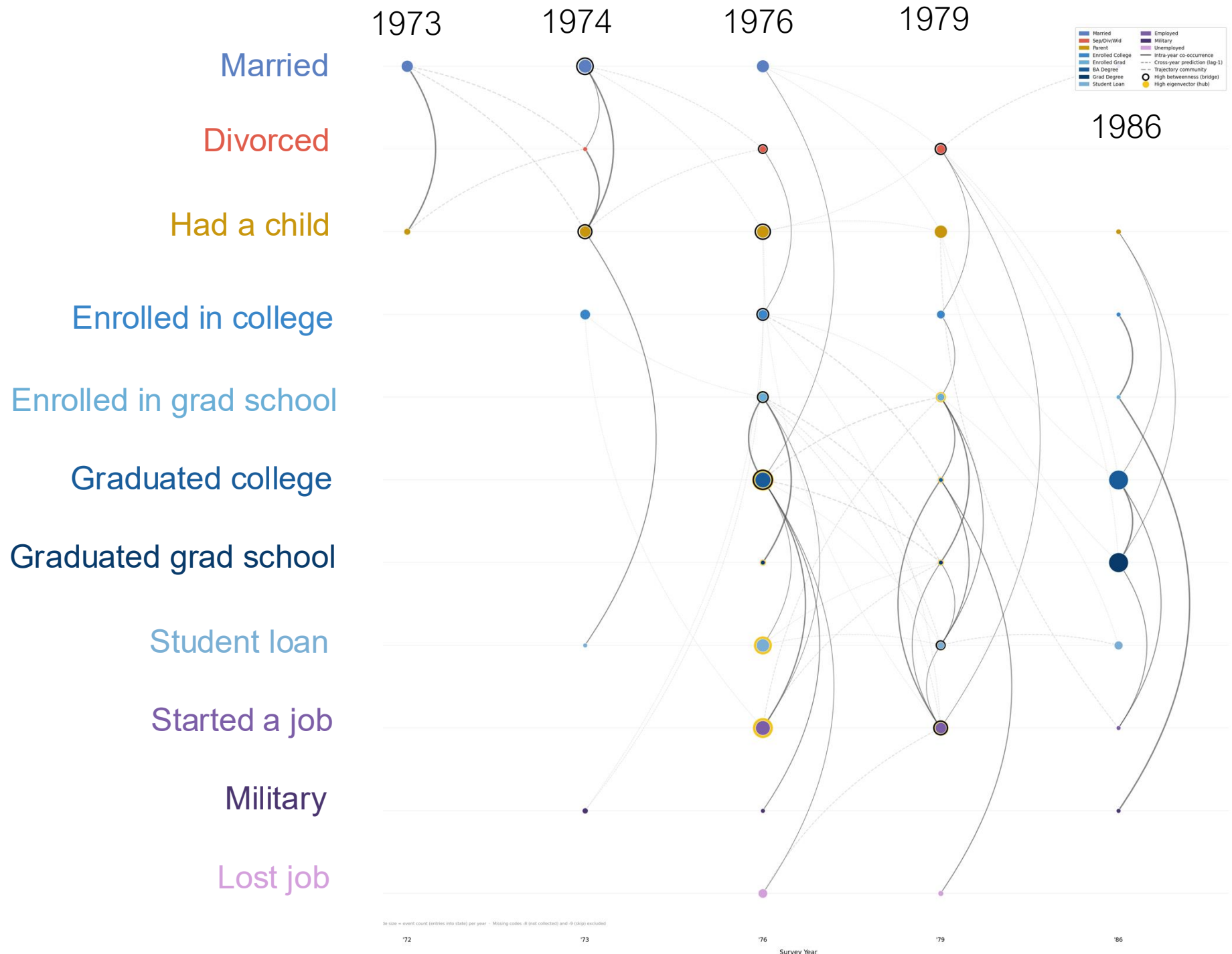


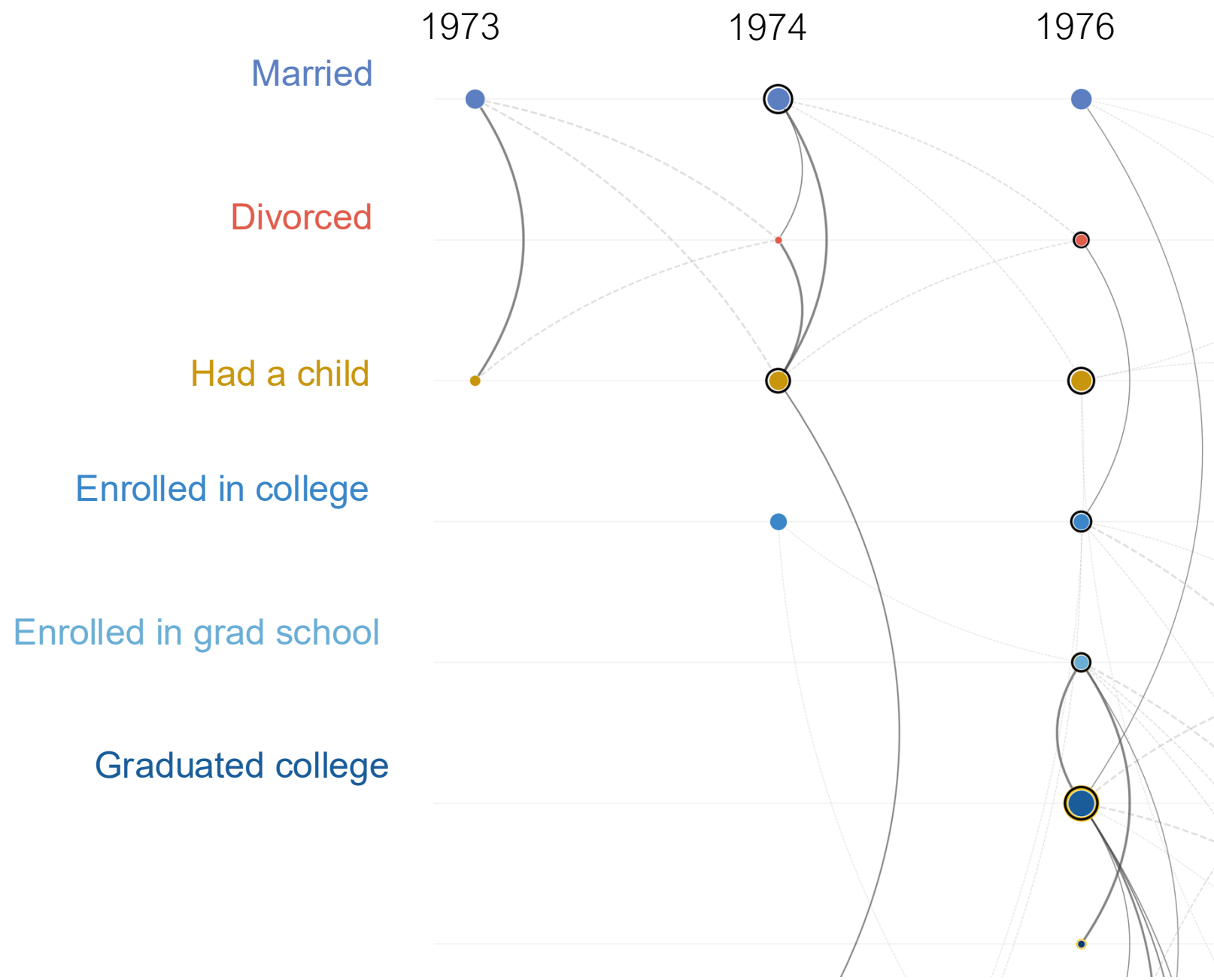
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**Arrows between years
connect specific life event
pairs across time.**

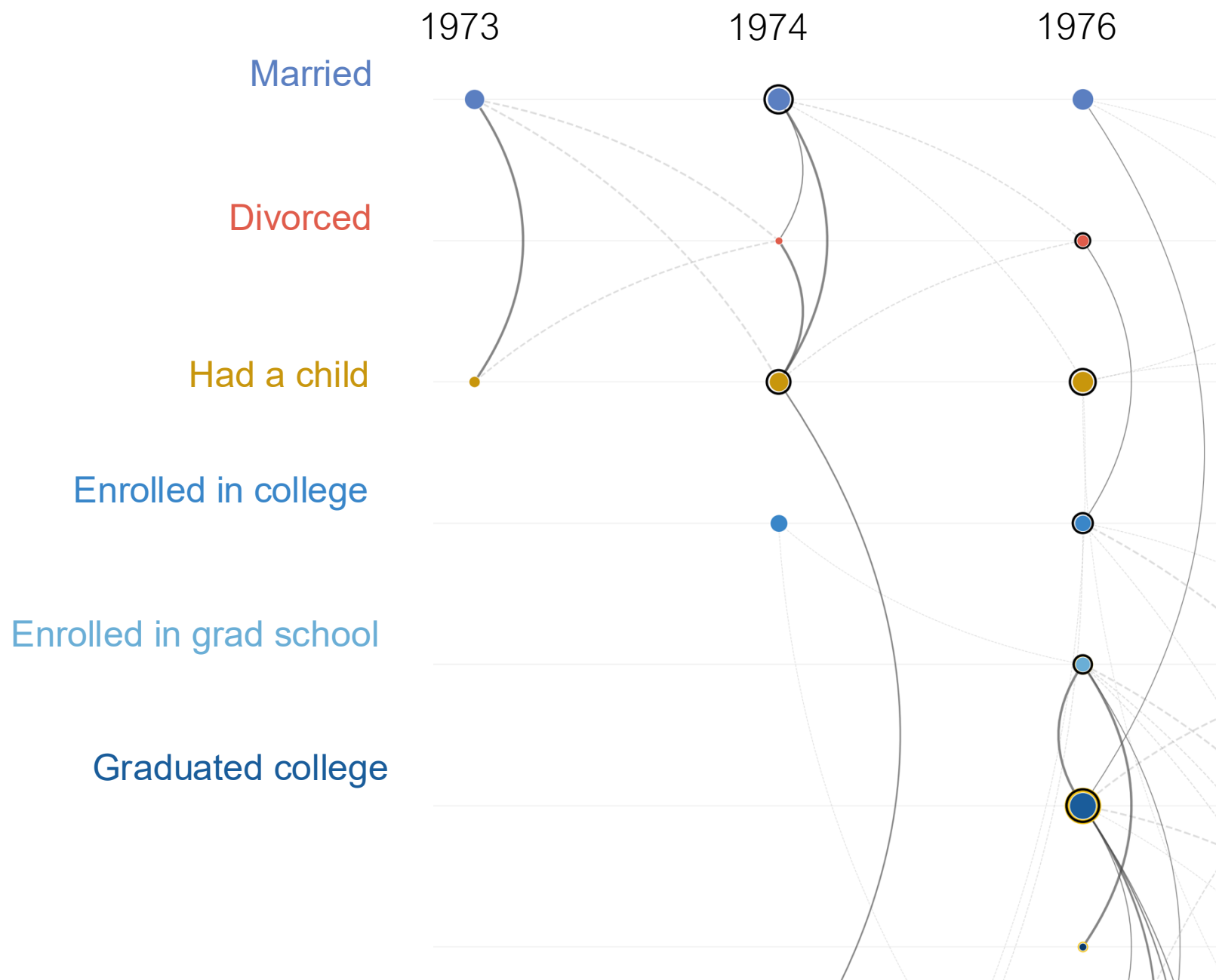
Lines within bins are
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High “betweenness” centrality = acts as a bridge between many other events





Marriage/parenthood early on is a structural bridge



Getting a degree is one of the most important structural bridges in the network, appears to reorganize trajectories more than family formation

Married

Divorced

Had a child

Enrolled in college

Enrolled in grad school

Graduated college

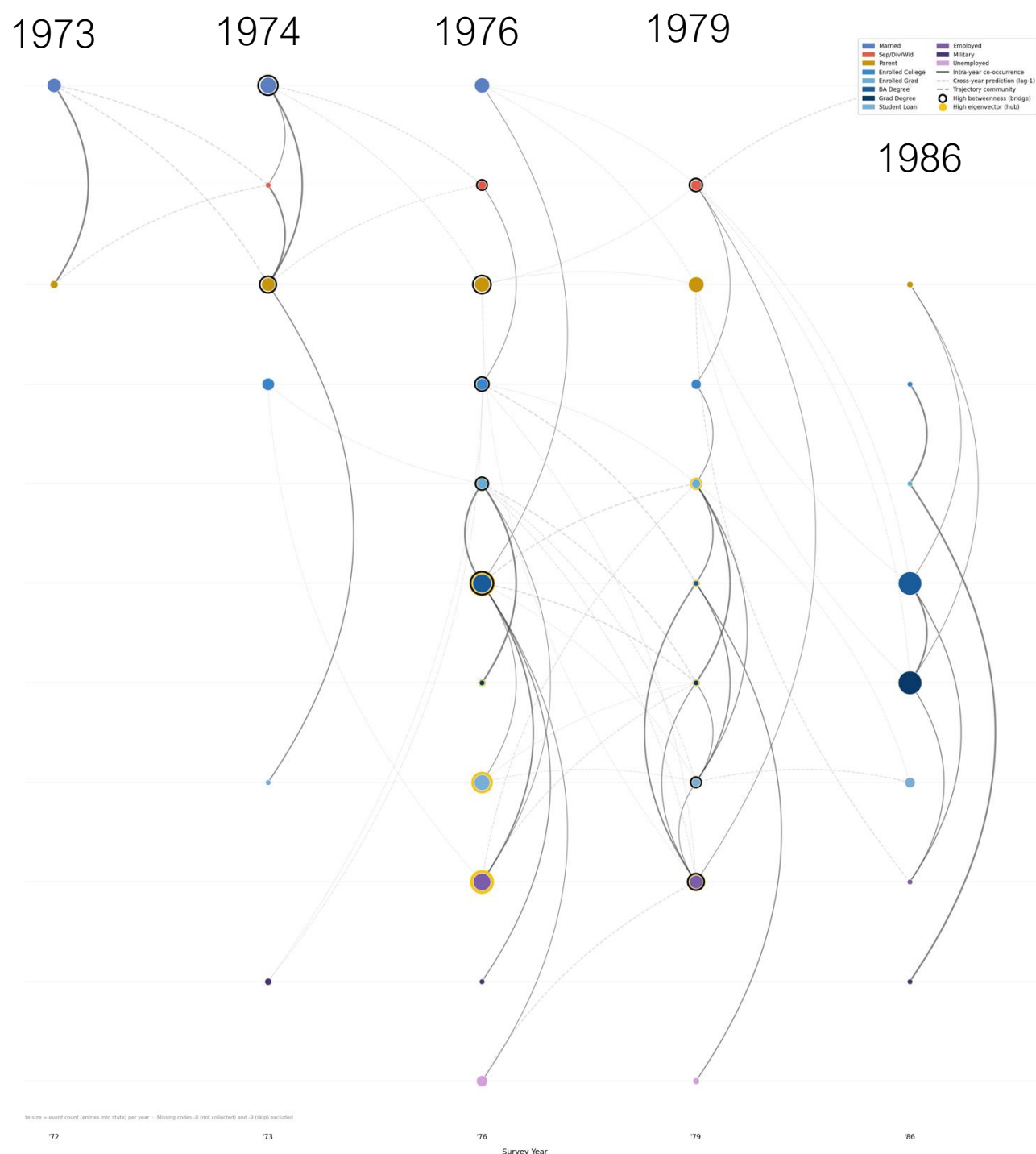
Graduated grad school

Student loan

Started a job

Military

Lost job



Far fewer cross-year connections after 1979, meaning much of the divergence occurs between ages 18-25

Summary

- Temporal co-occurrence networks reveal structure in life trajectories that traditional state-based methods miss

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Future Work

- Refine turning point definition – to what degree are edges in the following years are persistently reorganized compared to **matched peers**
- See how outcomes differ by turning points, like SES in 1986, and late-life outcomes from 2025!

Thank you



Tyler Matteson
Stanford



Bernardo Mackenna
Stanford



João Souto Maior
Stanford



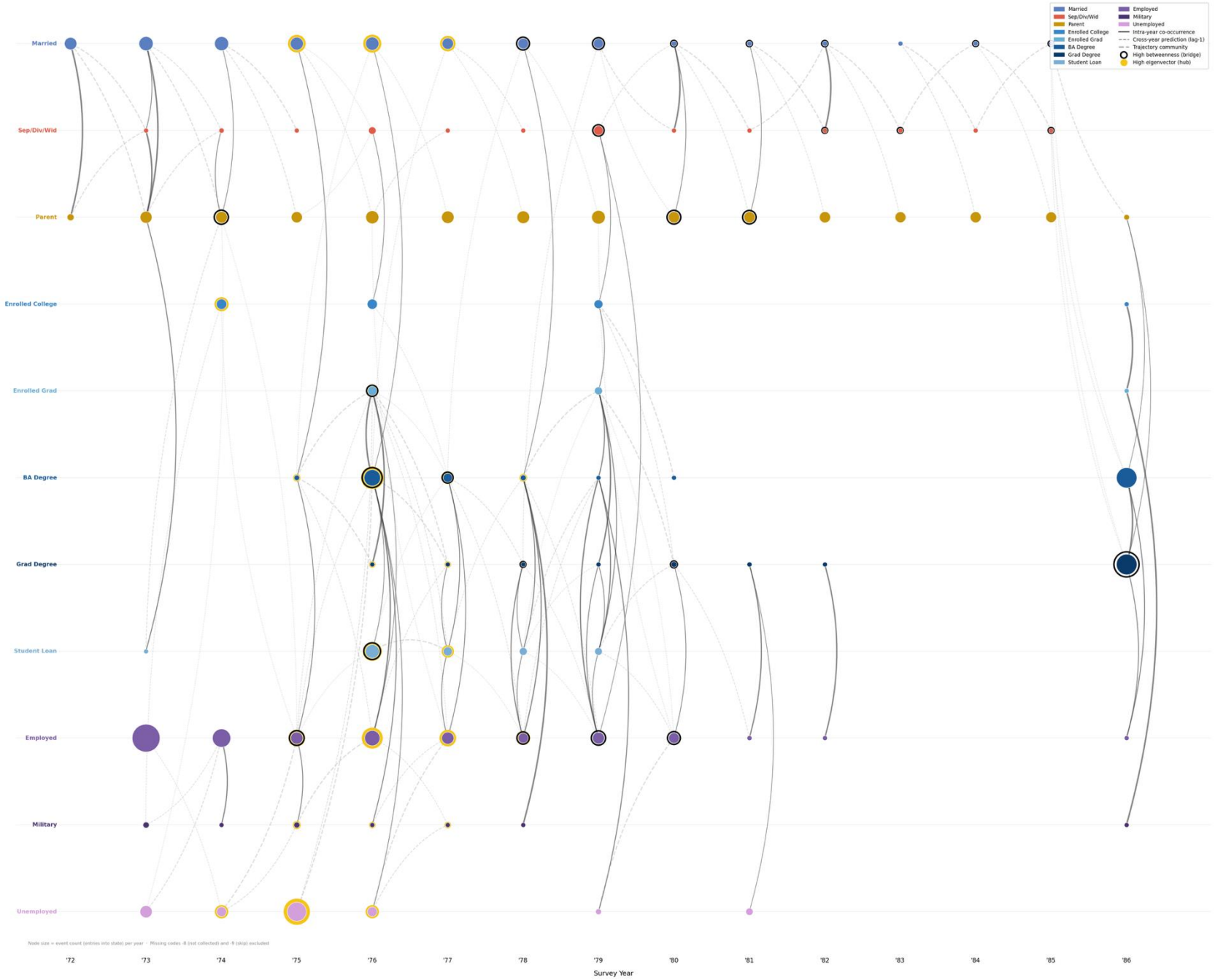
Phil Chodrow
Middlebury College



Carolina Chávez-Ruelas
Univ. Colorado Boulder



Carolina Pradier
Univ. Montreal



Networks –
can cluster
by
connections
instead of
similarity

But now we
can see
turning
points

